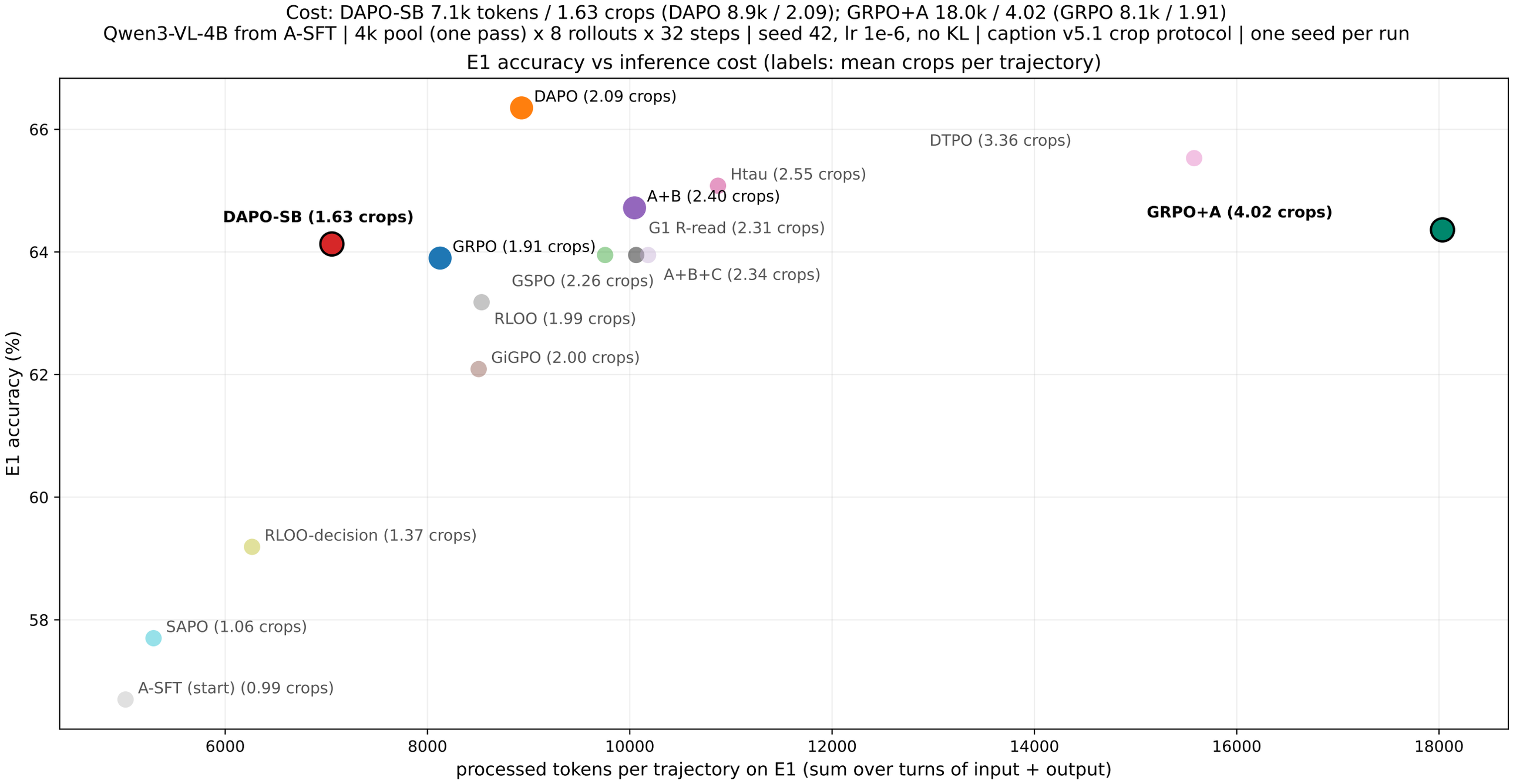
Cost: DAPO-SB 7.1k tokens / 1.63 crops (DAPO 8.9k / 2.09); GRPO+A 18.0k / 4.02 (GRPO 8.1k / 1.91)
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
E1 accuracy vs inference cost (labels: mean crops per trajectory)
DAPO (2.09 crops)
DTPO (3.36 crops)
Htau (2.55 crops)
A+B (2.40 crops)
DAPO-SB (1.63 crops)
GRPO+A (4.02 crops)
G1 R-read (2.31 crops)
GRPO (1.91 crops)
A+B+C (2.34 crops)
GSPO (2.26 crops)
RLOO (1.99 crops)
GiGPO (2.00 crops)
RLOO-decision (1.37 crops)
SAPO (1.06 crops)
A-SFT (start) (0.99 crops)
E1 accuracy (%)
66
64
62
60
58
6000
8000
10000
12000
14000
16000
18000
processed tokens per trajectory on E1 (sum over turns of input + output)